

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-27 09:18 UTC

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- Cycles recorded: **140**
- Time span: 83.6h (~1.67 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	121
FALSIFIED	14
SUPPORTED	4
BARRIER_HIT	1

Mathematical fields explored (field_A)

Field	Cycles
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier analysis of boolean functions	3
Noncommutative geometry	2
Representation theory of symmetric groups	2
Ergodic Circuit Framework (COMM_COMPLEXITY)	2
Matroid Theory	2
Algebraic Topology	2
Tropical geometry	2
Random Matrix Theory	2
Combinatorial homotopy theory	1
Kazhdan-Lusztig theory of Hecke algebras	1
Combinatorial algebraic topology	1
Knot theory (quantum invariants)	1
Quadratic forms over finite fields (Grothendieck-Witt groups)	1
Clifford algebras over finite fields	1
Toric geometry (via Gröbner degenerations)	1

Field	Cycles
Arithmetic geometry (Tate-Shafarevich groups)	1
Geometric Langlands correspondence	1
Algebraic graph theory (chromatic polynomials)	1
Clifford algebras over real vector spaces	1
Algebraic cycles (Chow groups over finite fields)	1
Algebraic lattice theory (Möbius functions of flow lattices)	1
Directed algebraic topology (d-space homology)	1
Motivic integration (Denef-Loeser zeta functions)	1
Matroid theory (Clifford index of binary matroids)	1
Representation theory of finite groups (Frobenius-Schur indicator)	1
Categorification and knot Floer homology	1
Modular forms	1
Algebraic geometry (ideals in polynomial rings)	1

Last 15 cycles (chronological)

Time	Verdict	Title
2026-04-27 01:38 UTC	INCONCLUSIVE	Pebble-Cost Gadget Lifts Decision Tree Depth to KW-Game Length
2026-04-27 02:14 UTC	INCONCLUSIVE	Mealy Automaticity Lower-Bounds Truth-Table Formula Size
2026-04-27 02:28 UTC	INCONCLUSIVE	Tropical Positive-Scalar Homogeneity of MinimalFourierCoefficient
2026-04-27 02:42 UTC	INCONCLUSIVE	Permanent-Variance of NW Design Matrix Predicts PRG Bias
2026-04-27 03:18 UTC	INCONCLUSIVE	Cycle-Space Cocycle Imbalance Lower-Bounds Resolution Width of Ts
2026-04-27 03:49 UTC	INCONCLUSIVE	Halton Star-Discrepancy of Clause-Sign Embedding Bounds DPLL Leav
2026-04-27 04:23 UTC	INCONCLUSIVE	Erdős-Ko-Rado Shadow Defect Lower-Bounds Disjointness Discrepancy
2026-04-27 04:57 UTC	INCONCLUSIVE	Edge-Expansion Defect Predicts Resolution Width of Tseitin on Ran
2026-04-27 06:23 UTC	INCONCLUSIVE	Average-Case DPLL Runtime Linked to Random Matrix Spectra
2026-04-27 07:04 UTC	INCONCLUSIVE	Betti Numbers and Resolution Width in 3-CNF
2026-04-27 07:15 UTC	INCONCLUSIVE	Sandpile-Group Order of Lifted IP Bipartite Graph Bounds Decision

Time	Verdict	Title
2026-04-27 07:44 UTC	INCONCLUSIVE	Doob Martingale Variance Gap Predicts Worst-Case DPLL Depth
2026-04-27 08:18 UTC	INCONCLUSIVE	Newton-Inequality Defect of SAT Generating Polynomial Bounds DPLL
2026-04-27 08:47 UTC	INCONCLUSIVE	Möbius Function of Minterm Lattice Bounds Indexing-Lifted Communi
2026-04-27 09:18 UTC	INCONCLUSIVE	Noise-Stability Plateau at $\rho=1/3$ Bounds DPLL Leaves on 3-CNF

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```